Freshwater

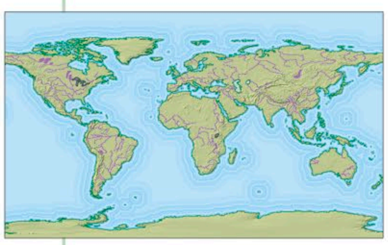
Every year, about 24,000 cubic miles (100,000 cubic km) of water evaporates from the world's oceans, condenses, and then falls as rain or snow. Most of this water disappears back into the atmosphere to continue this cycle, but about a third returns to the oceans by flowing either over ground or beneath the land surface. This steady supply of freshwater sustains all the world's land-based life, as well as creating highly diverse habitats—from streams, rivers, and lakes to reedbeds, marshes, and swamps—in which a wide range of different animal and plant life can thrive.

Lakes and rivers

For permanent water dwellers, life in lakes and rivers is shaped by many different factors. One of these is the water's chemical makeup, which is often dictated by the type of rock that forms the bed of a river or lake. Hard water, for example, is good for animals that grow shells because it contains calcium that can be used as shell-building material, while water that is especially rich in oxygen is important for highly active predatory fish, such as salmon and trout. Water that is very deficient in oxygen, on the other hand, provides a poor environment for animal life as a whole because relatively few aquatic species, other than specialized worms, can survive in it.

In lakes and rivers, aquatic animals usually occupy clearly defined zones. The brightly lit water close to the surface often teems with water-fleas, copepods, and other microscopic forms of animal life. They live here in order to feed on phytoplankton—the microscopic algae that become extremely abundant during the summer months. Feeding in the middle zone and near the water's surface are larger animals, such as fish, that are able to hold their own against the strength of the current. Weak swimmers live near the banks, where the current is slower, or among stones and sediment on the bottom. In still and slow-flowing water, surface tension supports insects that hunt by walking or running over the water.

Some animals that are associated with freshwater habitats are not necessarily permanent water dwellers; instead, they divide their time between water and the adjacent land, entering the lake or river to hunt or breed, or using it as a nursery for their young.



IN GEOLOGICAL TERMS, *lakes and rivers are highly changeable. Lakes gradually fill with sediment, while rivers often change course.*

HERONS *hunt by stealth, waiting along riverbanks and lake shores for fish to come within range.*

DRAGONFLIES, *like many other insects, spend the early part of their lives in freshwater, leaving it only when they are ready to breed.*

FROGS *vary in their affinity for water. The American bullfrog rarely leaves it, even as an adult.*

MANY AMPHIBIANS, *such as the great crested newt, use still or slowly flowing water containing plenty of plants as a repository for their eggs.*

SMALL CRUSTACEANS *such as this cyclops copepod, are part of the animal plankton in ponds and lakes.*

PIKE *are sit-and-wait predators, using dense waterside vegetation as cover from which to ambush their prey.*